Q1

4.4 Find the values of I and V in the circuits shown in Fig. E4.4.

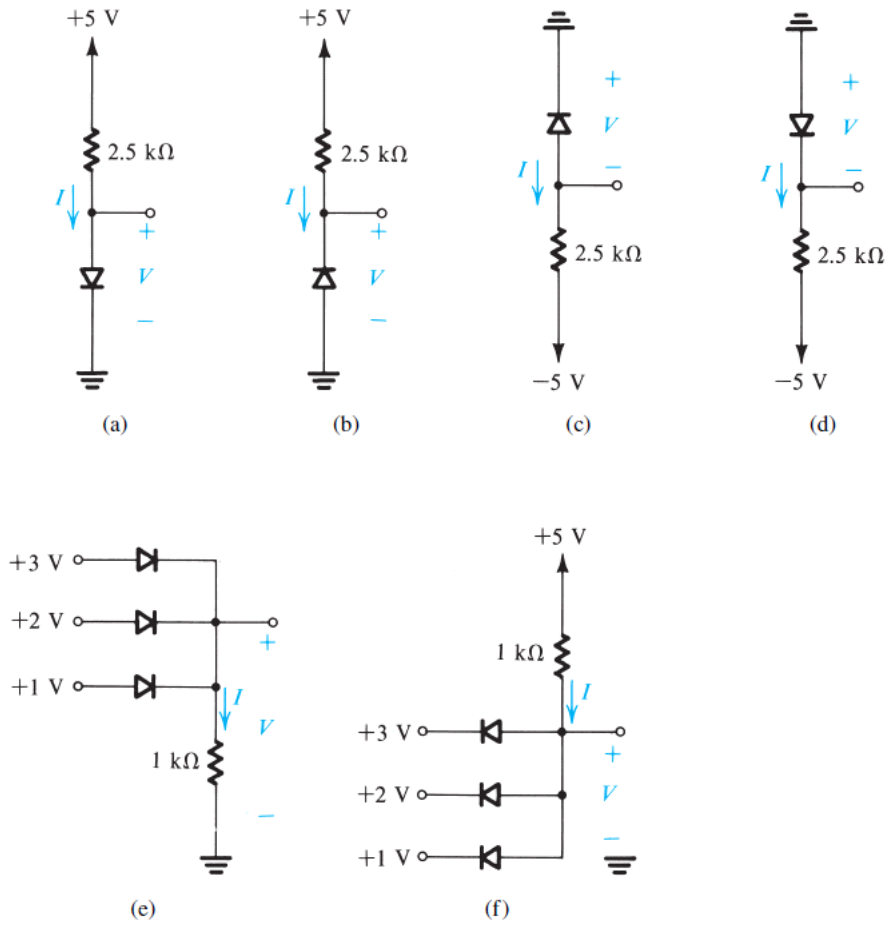


Figure E4.4

Q2

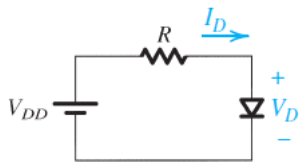


Figure 4.10

For the circuit in Fig. 4.10, find I_D and V_D for the case $V_{DD} = 5$ V and $R = 10$ kΩ. Assume that the diode has the constant-voltage-drop model with $V_D = 0.7$ V.

Ans. 0.43 mA, 0.7 V

Q3

A zener diode whose nominal voltage is 10 V at 10 mA has an incremental resistance of 50 Ω. What voltage do you expect if the diode current is halved? Doubled? What is the value of V_{Z0} in the zener model?

Ans. 9.75 V; 10.5 V; 9.5 V